MILITARY SPECIFICATION

VIEWER, STILL PICTURE AR-12( )

This specification is approved for use by all the Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of portable viewer for examining various types of transparent films, designated Viewer, Still Picture AR-12( ).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods Of.
MIL-W-6858	- Welding, Resistance, Spot And Seam.
MIL-C-10578	- Corrosion Removing And Metal Conditioning Compound (Phosphoric Acid Base).
MIL-M-1323	- Marking Of Electronic Items.
MIL-F-14072	- Finishes For Ground Electronic Equipment.

STANDARDS

MILITARY

MIL-STD-22	- Welded Joint Design.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, NJ 07703-5304, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6730

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MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-252	- Wired Equipment, Classification Of Visual And Mechanical Defects.
MIL-STD-454	- Standard General Requirements For Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement Of.
MIL-STD-810	- Environmental Test Methods.

2.1.2 Other Government documents. The following Government documents form a part of this specification to the extent specified herein:

DRAWINGS

ELECTRONICS COMMAND

SC-DL-74583	- Viewer, Still Picture AR-12(1).
SPI 1G00080	- Special Packaging Instructions.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The viewer is a portable table-top unit containing a light source and a flashed opal glass designed to permit examination of transparencies, negatives, and X-ray film up to 10 by 10 inches (25.4 by 25.4cm) in size. The opal glass vieweing surface is set at a slight angle with the vertical to facilitate use of the unit.

3.2 First article. When specified (see 6.2) a sample shall be otherwise specified in the contract, the sample units shall be subjected to first article inspection (see 4.3, 6.2b).

3.3 Material. The material shall be as specified herein and as shown on the applicable drawings. However, when a definite material is not specified, a material shall be used which will enable the viewer to meet the performance requirements of this specification. Acceptance of approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Recycled material. It is encouraged that recycled material be used when practical provided it meets the requirements of this specification.

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3.3.2 Cleaning. Cleaning of all material, parts and the assembled viewer shall be in accordance with requirement 9 of MIL-STD-454.

3.4 Construction. The viewer shall be constructed in accordance with drawing and data list SC-DL-74583.

3.4.1 Welding. Wherever practicable, welded joints shall conform to MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to MIL-W-6858. Spot or other intermittent welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. When spot or other intermittent welds are used to hold a part, the number of welds shall be not less than two.

3.4.1.1 Cleaning prior to welding. Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.4.1.2 Process. Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas shielded arc welding shall be used, when practicable, for welding of aluminum, magnesium, or stainless steel.

3.4.1.3 Cleaning after welding. Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with MIL-C-10578. Acid used for cleaning shall be completely neutralized and removed.

3.4.2 Electrical system.

3.4.2.1 Power supply. The viewer shall be operable from a nominal 115 volt, 60 Hertz (Hz), alternating current (ac) power source.

3.4.2.2 Wiring. Electrical wiring shall be in accordance with best commercial practice. Size and location of wire, insulation and switch-mountings shall be suitable for the purpose intended and shall be adequate to meet the tests specified herein. Wire slack shall conform to MIL-STD-454 requirement 9 and shall provide for at least two replacements of the part to which the wire is connected.

3.4.2.3 Breakdown voltage, insulation. Electrical wiring shall be subjected to a breakdown voltage of 650 (+10 or -0) volts for not less than 10 seconds. After the 10 second application of the voltage, there shall be no visible deterioration of the insulation and no degradation of the illumination and operational performance specified in 3.5 and 3.5.1.

3.5 Operational suitability. When tested as specified in 4.6.2, film negatives shall not be scratched when inserted in the viewing position and then removed and there shall be no emission of stray light.

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3.5.1 Illumination. When tested as specified in 4.6.3, the viewer shall give a reading of 125 ±20 footcandle (1345 ±215 lux) over the entire viewing surface.

3.6 Environmental. The viewer shall withstand individual environmental conditions specified herein, or cominbations thereof without visible damage, degradation of operation (see 3.5), degradation of illumination (see 3.5.1), or breakdown of wiring insulation (see 3.4.2.3) when tested as specified in 4.6.4.

3.6.1 Low temperature. Before, during, and after testing as specified in 4.6.4.1.1, the equipment shall meet full specified requirements as specified in 3.5 and 3.5.1.

3.6.2 High temperature. Before, during, and after testing as specified in 4.6.4.1.2, the equipment shall meet full specified requirements as specified in 3.5 and 3.5.1.

3.6.3 Humidity (moisture-resistance). During testing as specified in 4.6.4.1.3, the equipment shall meet full specified requirements as specified in 3.4.2.3. In addition, after the test of 4.6.4.1.3, the equipment shall exhibit no physical damage, such as rust or other corrosion, blistering, swelling or deterioration of parts or materials.

3.7 Electromagnetic interference characteristics. The viewer shall meet the electromagnetic interference requirements specified in table 1-11, class C3, of MIL-STD-461. If the contractor considers it desirable to modify the viewer in order to facilitate conformance to the specified test limits, he shall obtain approval from the contracting officer before proceeding with the modification.

3.8 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.9 Finish. The viewer shall be finished in accordance with MIL-F-14072, and as specified on the drawings.

3.10 Marking. Marking shall conform to MIL-M-13231 and the applicable equipment drawings.

3.11 Serial numbers. Each Viewer, Still Picture AR-12( ) shall be serial numbered.

3.12 Workmanship. The viewer shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs herein:

3.3.2 Cleaning.
3.4.1.1 Cleaning prior to welding.
3.4.1.2 Process for welding.
3.4.1.3 Cleaning after welding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classifications of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Packaging inspection (see 4.7).

4.3 First article.

4.3.1 First article units. When required and as specified in the contract, the contractor shall furnish first article units of the complete viewer.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, and the inspections specified for group A, group B and group C. The inspection shall be performed in the following order: 4.4, group A and group B and group C. After completion of group C environmental tests, conforming units shall be reinspected and shall pass all group A inspections. Failure of any inspection shall be cause for rejection.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification.

Item	Where required
Finish	3.9
Marking	3.10

4.5 Quality conformance inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspections specified in table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined, or as specified herein, shall be in accordance with MIL-STD-105, using the general inspection levels and acceptable quality levels (AQL) specified in table I.

TABLE I. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	AQL (percent defective) Major	Minor
Visual and mechanical	3.12	4.6.8	1.0	4.0
Operational	3.5	4.6.2	1.0	1/
Illumination	3.5.1	4.6.3	1.0	1/
Interchangeability	3.8	4.6.7	1.0	4.0

1/ All defects found in the illumination and operational inspection shall be considered major.

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4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in any order satisfactory to the Government except that the operational inspection (4.6.2) shall be performed last.

4.5.2 Group B inspection. This inspection shall consist of the inspection specified for breakdown voltage in 4.6.1 and shall conform to the special procedures for small-sample inspection of MIL-STD-105. The AQL shall be 6.5 percent defective, the inspection level shall be S-4 for normal and tightened inspection, and S-3 for reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table II.

TABLE II. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Low temperature	3.6.1, 3.6.2	4.6.4.1.1
High temperature	3.6.1	4.6.4.1.2
Humidity	3.6.3	4.6.4.1.3
Electromagnetic interference	3.7	4.6.6

4.5.3.1 Sampling for group C inspection. One sample unit shall be selected at random from every 50 units, or fraction thereof, produced. The first sample selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Group C failures. Actions required relative to group C failure shall be as specified in the contract (see 6.2).

4.6 Methods of inspection.

4.6.1 Voltage breakdown.

4.6.1.1 Voltage. The applied voltage shall be 650 (+10 or -0) volts of either direct current or peak ac. If ac, the waveform shall be approximately sinusoidal and the frequency not more than 100 Hz.

4.6.1.2 Procedure. With the switch in the OFF position, apply the voltage across the switch terminals and between one uninsulated portion of each wire and ground for a period of not less than 10 seconds. At the conclusion of testing, inspect the insulation for visual deterioration and test the viewer as specified in 4.6 except that portion of 4.6 involving insertion of a film negative in the viewing position is eliminated. Nonconformance to 3.4.2.3 shall constitute failure of this test.

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4.6.2 Operational test. A film negative shall be inserted in the viewing position against the opal glass with use of all the film clips. Then the film negative shall be removed. This operation of insertion and removal of negative shall be repeated five times. All film clips shall be observed for proper operation and for non-scratching of the negative. The line switch shall be operated ten times, and observation made for the presence of positive contact as indicated by the light going on and off. Examine for emision of stray white light during this test. Nonconformance to 3.5 shall constitute failure of this test.

4.6.3 Illumination. The viewer shall be connected to a variable transformer, with the voltage adjusted to 115 ±1 volts and the bulb lighted. A Weston illumination meter, P/N 268560, or equal, shall be placed in contact with the viewing surface. Readings shall be taken over the entire viewing area. Nonconformance to 3.5.1 shall constitute failure of this test.

4.6.4 Environmental.

4.6.4.1 Temperature.

4.6.4.1.1 Low temperature. The viewer shall be tested in accordance with method 502.1, procedure I of MIL-STD-810. The storage temperature shall be -60 (+0 or -6)°F (-51 (+ or -3.3)°C) and the low operating temperature shall be 32 (+0 or -6)°F ((+0 or -3.3)°C). A temperature of -60°F (-51°C) shall be maintained until equilibrium is reached, but shall not exceed 6 hours.

4.6.4.1.2 High temperature. The viewer shall be tested in accordance with method 501.1, procedure II, of MIL-STD-810. The high operating temperature shall be 125 (+6 or -0)°F (52 (+3.3 or -0)°C).

4.6.4.1.3 Humidity (moisture-resistance). The viewer shall be subjected to humidity-temperature conditions in accordance with method 507.1, procedure II of MIL-STD-810. The viewer shall be tested as specified in 4.6.2 at the following times:

a. At the end of step 3 of procedure II (after exposure to 23°C (73.4°F) and 50 ±10 percent relative humidity for 24 hours). The ambient conditions of step 3 shall be maintained while the operational test is in progress.

b. Anytime during the last hour of the final 48-hour cycle of step 6 of procedure II. (While the operational test is in progress, nominal ambient conditions will be 30°C (86°F) and 94 percent relative humidity.

4.6.5 Failure criteria. Nonconformance to 3.6 shall constitute failure of environmental tests.

4.6.6. Electromagnetic interference test. The viewer shall be tested for electromagnetic interference characteristics in accordance with MIL-STD-462. Nonconformance to 3.7 shall constitute failure of this test.

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4.6.7 Inspection for interchangeability. The dimensions shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.8. When a listed dimension is not within specified limits, it shall be considered a major defect.

4.6.8 Visual and mechanical. The viewer shall be examined for the defects listed in MIL-STD-252. Nonconformance to 3.12 shall constitute failure of this test.

4.7 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.7.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.7.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.7.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table III "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and an acceptable quality level of 4.0 percent defective.

TABLE III. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, or omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI 1G00080.

6. NOTES

6.1 Intended use. The viewer is a portable table-top unit used for examination of transparencies, negatives, and x-ray film.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. When a first article is required for inspection and approval.
c. Group C failure action (see 4.5.3.2).
d. Level A or B preservation and packing (see section 5).
e. When first article inspection rough handling tests are not required.
f. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article model. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AR-12W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.5 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER

Preparing activity:
Army - ER
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-V-118158(ER)	VIEWER, STILL PICTURE AR-12( )

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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